

Excess kurtosis of the even-monomial Gibbs weight

Tide retrospective, laplace seabed

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1 Setting

The 1D monomial track of the laplace formalisation computes moments of x against the symmetric reference weight $\exp(-t x^{2k}/(2k)!)$ in Gamma closed form. The second cumulant — the variance — was already landed as `monomial_variance_even`. This tide adds the fourth cumulant, the connected four-point function, which for a symmetric weight (mean zero) is $\langle x^4 \rangle - 3\langle x^2 \rangle^2$. It is the minimal good next step: a self-contained closed form that reuses only machinery already present, and mirrors an already-proved lemma structurally. It also opens the cumulant ladder past the variance on the monomial track, complementing the anharmonic track's κ_3, κ_4 work.

2 The thing formalised

For $k \geq 1$ and $t > 0$,

$$\langle x^4 \rangle - 3\langle x^2 \rangle^2 = \left(\frac{(2k)!}{t} \right)^{2/k} \left(\frac{\Gamma(5/(2k))}{\Gamma(1/(2k))} - 3 \left(\frac{\Gamma(3/(2k))}{\Gamma(1/(2k))} \right)^2 \right).$$

The Lean statement is `monomial_excess_kurtosis`, phrased with the raw `gibbsExpectation` of x^4 and x^2 . For the Gaussian reference $k = 1$ the bracket is $\frac{3}{4} - 3 \cdot (\frac{1}{2})^2 = 0$: a Gaussian has zero excess kurtosis, the sanity check the tide began from.

3 Proof strategy

The proof is a near-verbatim structural copy of `monomial_variance_even`. Rewrite x^4 and x^2 into the even form $x^{2 \cdot 2}$, $x^{2 \cdot 1}$ (definitionally equal, so `rfl`), then substitute the even-moment closed form `gibbsExpectation_kthPotential_even` at $j = 2$ and $j = 1$. The two terms then share the power $((2k)!/t)^{2/k}$, once the square of the $j = 1$ term's power $((2k)!/t)^{1/k}$ is folded in via $((2k)!/t)^{2/k} = (((2k)!/t)^{1/k})^2$, an instance of `Real.rpow_add`. After a `push_cast` to align the natural-number casts in the exponents and Gamma arguments, that single power rewrite plus `ring` closes the goal — the Gamma ratios ride along as opaque atoms.

4 Roadblocks and resolutions

None. The tide compiled on the first attempt; the mirror discipline (find the closest landed lemma, copy its skeleton, change the arithmetic) made it a mechanical exercise once the survey had picked the target. The only care needed was ordering `push_cast` before the power rewrite so the $\mathbb{N}$ -cast 2 in the substituted exponent matches the $\mathbb{R}$ -literal 2 in the `Real.rpow_add` lemma.

5 What was Mathlib, what was new

Everything analytic was already in the seabed: the closed-form even moments, the integrability inputs behind them, and the `Real.rpow` algebra. New here is the fourth-cumulant closed form itself, and the observation that it vanishes at $k = 1$. No GPT consult was needed.